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Unit – 4 : Session – 7 : SLO – 1 | Generation of Control Signals — Fill in the Blanks

■ I. Fill in the Blanks

1. opcode — control signals are generated based on the opcode field of the instruction.
2. control (decoder) unit — decodes the instruction to determine which signals to activate.
3. control memory (control store) — in microprogrammed control, the control signals are stored as microinstructions in control memory.
4. sequence counter — tracks the current step in the instruction cycle.
5. Hardwired — hardwired control uses combinational logic to generate signals directly.
6. MEMREAD (memory read) — the signal that tells the processor to read from memory.
7. opcode / function-code — ALU control signals are determined by the opcode (function field) input to the ALU control.
8. Instruction Register (IR) — holds the current instruction being executed.
9. individual bits (one bit per signal) — horizontal microprogramming represents each control signal as its own bit in a wide microinstruction word.
10. MEMWRITE (memory write) — initiates a write to memory or an I/O device.